

Yufan Du

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Education

Peking University **Sep. 2021 - Present**
Bachelor of Science in Applied Physics *and* Computer Science Dual Major *Beijing, China*

Relevant Scores

Overall GPA: 3.847/4.0 (top 10% of 66 students)
TOEFL: 108(R27 + L30 + S23 + W28)
GRE: 324 (Quantitative + Verbal), 4.0 (Analytical Writing)

Technical Skills

Knowledgeable with: C/C++, Python
Familiar with: Python ML libraries (PyTorch, NumPy), Verilog, Embedded systems, FPGA, SQL/Database, JavaScript/HTML, Linux systems, MATLAB, Tcl, L^AT_EX
EDA tools: Familiar with Cadence Innovus, Cadence Virtuoso

Publications

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- **Yufan Du**[†], Zizheng Guo[†], Xun Jiang, Zhuomin Chai, Yuxiang Zhao, Yibo Lin, Runsheng Wang and Ru Huang, "PowPrediCT: Cross-Stage Power Prediction with Circuit-Transformation-Aware Learning," *2024 Design Automation Conference (DAC24)*.
 - **Yufan Du**, Zizheng Guo, Yibo Lin, Runsheng Wang and Ru Huang, "Fusion of Global Placement and Gate Sizing with Differentiable Optimization," *2024 International Conference on Computer-Aided Design (ICCAD24)*. (**best paper candidate in track**).
 - Zizheng Guo, Zuodong Zhang, Wuxi Li, Tsung-Wei Huang, Xizhe Shi, **Yufan Du**, Yibo Lin, Runsheng Wang and Ru Huang, "HeteroExcept: A CPU-GPU Heterogeneous Algorithm to Accelerate Exception-aware Static Timing Analysis," *2024 International Conference on Computer-Aided Design (ICCAD24)*.

Project Experiences

FPGA-Based Snake Game Design Project **Mar. 2023 - Jun. 2023**
- Advisor: Professor Xiaohui Duan.
- Deployed a computer vision algorithm and the classic Snake game on a resource-limited FPGA platform, optimizing resource usage.
- Developed and implemented a real-time control mechanism for the game using hand movement detection with a camera, enhancing user interaction.
- Introduced a 3D human-computer interface to improve player engagement and provided valuable insights for IoT edge device development.

AI-Assisted Schedule Manager **May. 2023 - July. 2023**
- Advisor: Professor Wei Guo.
- Developed a cross-platform scheduling application integrated with AI, applying NLP for user-friendly schedule management with commercial API.

- Achieved a substantial improvement in user experience by enabling efficient management and reminders through conversational interfaces.

Research Experiences

Cross-Stage Power Prediction for Integrated Circuits Jun. 2023. - Nov. 2023.

- Advisor: Professor Yibo Lin.
- Served as the first author.
- Proposed an innovative VLSI circuit power prediction framework that integrates cross-stage circuit-transformation-aware learning.
- Achieved a significant reduction in error rates and computation time compared to industry-leading commercial tools.
- It thus enabled effective power optimization strategies in early-stage VLSI design.
- *Accepted by DAC 2024.*

CUDA-Accelerated Gate Sizing Dec. 2023 - May. 2024

- Advisor: Professor Yibo Lin.
- Served as the first author.
- Proposed a novel fusion of global placement and gate sizing with differentiable optimization that could be accelerated by GPU.
- Unlocked a broader space for optimization and provided a guideline for optimization algorithms.
- Demonstrated a substantial improvement in runtime efficiency and PPA metrics compared to traditional CPU-based methods.
- *Best paper candidate of track in ICCAD 2024.*
- *Accepted by ICCAD 2024.*